Verification of LESCO Residential Electricity Tariffs, Billing Rules, Prosumer Net-Billing Framework, and Metering Infrastructure in Lahore (CY 2026)

Source-Backed Regulatory Audit for Solar & Battery Energy-Management System (EMS) Configuration | Effective Tariff Regime: S.R.O. 46(I)/2026, S.R.O. 251(I)/2026, S.R.O. 279(I)/2026, S.R.O. 547(I)/2026, S.R.O. 1330(I)/2026, S.R.O. 1501(I)/2026 & S.R.O. 1754(I)/2026 (As of October 2026)

Executive Regulatory Overview and Calendar-Year Rebasing Architecture

The residential retail electricity tariff structure governing Lahore Electric Supply Company (LESCO) underwent two foundational structural transformations during the first quarter of calendar year 2026. First, the National Electric Power Regulatory Authority (NEPRA) transitioned the national tariff determination cycle from the historical fiscal-year schedule (July to June) to a calendar-year rebasing framework effective 1 January 2026. On 7 January 2026, the Authority issued its comprehensive determination for LESCO's revised annual rebasing for Calendar Year 2026 (Case No. NEPRA/TRF-603 & TRF-604, Decision dated 07-01-2026) alongside the National Power Purchase Price (PPP) forecast determination for the Central Power Purchasing Agency (Case No. NEPRA/TRF-100/CPPAG, Decision dated 07-01-2026). Following a motion filed by the Federal Government under Sections 7 and 31 of the NEPRA Act, NEPRA approved the Uniform Consumer-End Tariff on 12 January 2026 (Decision on Motion of Federal Govt. No. 1004-23 dated 12-01-2026), which the Ministry of Energy (Power Division) notified in the Official Gazette on 13 January 2026 via Statutory Regulatory Orders S.R.O. 41(I)/2026 through S.R.O. 52(I)/2026, with S.R.O. 46(I)/2026 specifically governing LESCO.

Second, less than one month into the new calendar-year schedule, the Federal Government filed a tariff rationalization motion designed to reduce the cross-subsidy burden borne by industrial consumers (approximately Rs 131 billion nationwide) by restructuring residential fixed charges and slightly trimming upper-slab residential variable rates. NEPRA approved this cross-subsidy rationalization on 11 February 2026 (NEPRA Tariff Rationalization Decision dated 11-02-2026), and the Power Division notified the revised Schedule of Tariff on 12 February 2026 via S.R.O. 279(I)/2026 (within the S.R.O. 274(I)/2026 to S.R.O. 284(I)/2026 series), as corroborated by Business Recorder (12-02-2026) and The Express Tribune (14-02-2026). Consequently, the base Schedule of Tariff established in July 2025 under S.R.O. 1161(I)/2025 dated 01-07-2025 and the initial January 2026 notification (S.R.O. 46(I)/2026) have been superseded in respect of residential fixed charges, unprotected energy charges above 300 units, and Time-of-Use (ToU) off-peak energy charges.

Simultaneously, on 9 February 2026, NEPRA promulgated the NEPRA (Prosumer) Regulations, 2026 via S.R.O. 251(I)/2026, repealing the decade-old 2015 Net Metering Regulations and replacing one-to-one unit netting with gross net-billing at the National Average Energy Purchase Price (NAEPP) for new distributed generation applicants, before issuing a critical grandfathering amendment via S.R.O. 547(I)/2026 dated 02-04-2026 and a 25 kW concurrence exemption via S.R.O. 1330(I)/2026 dated 06-08-2026. For any solar and battery energy-management system (EMS) operating in Lahore, these overlapping tariff and regulatory shifts fundamentally alter dispatch optimization, peak-shaving economics, and self-consumption vs. grid-export valuation.

Explanation of Primary vs. Secondary Source Discrepancies Across Pakistani Tariff Literature

A recurring challenge when auditing Pakistani electricity tariff data is that primary regulatory determinations and secondary consumer portals frequently publish seemingly divergent figures for the same residential

slab. Rigorous cross-examination of primary NEPRA schedules reveals four systematic reasons for these discrepancies:

1. **NEPRA Determined Uniform Tariff vs. Government-Notified Applicable Tariff**: Every Schedule of Tariff (SoT) contains two parallel sets of columns: the *NEPRA Determined Uniform Tariff* (reflecting full cost-of-service recovery with and without Prior Year Adjustments, which in the 12-01-2026 Uniform Tariff Decision stands at Rs 36.13/kWh across all slabs up to 300 units) and the *Government of Pakistan (GoP) Applicable Uniform Tariff* (the subsidized or cross-subsidized rate actually billed to consumers after applying Tariff Differential Subsidies). Furthermore, LESCO's standalone cost-of-service determination (07-01-2026 LESCO Rebasing Decision, Annex-III, p. 17) lists LESCO-specific cost-of-service rates (e.g., Rs 31.14/kWh across slabs up to 300 units, Rs 28.01/kWh Peak, and Rs 25.62/kWh Off-Peak) prior to national uniformization. Consumer billing calculators use the GoP Applicable column, whereas regulatory analyses of utility revenue requirements cite the NEPRA Determined column. Both values are presented side-by-side throughout this report.
2. **Superseded Notifications (July 2025 vs. January 2026 vs. February 2026)**: Several utility bill calculators and blogs continue to display the July 2025 Schedule of Tariff (S.R.O. 1161(I)/2025 dated 01-07-2025), which listed higher variable charges for unprotected slabs above 300 units (e.g., Rs 37.99/kWh for 301–400 units and Rs 47.69/kWh above 700 units), an off-peak ToU rate of Rs 40.53/kWh, and per-connection fixed charges only above 300 units. Those rates were superseded on 12 February 2026 by S.R.O. 279(I)/2026.
3. **Base Energy Rate vs. Effective All-In Rate**: News headlines and consumer guides frequently quote an "effective per-unit rate" that bundles the base GoP variable energy charge with the Financing Cost (FC) Surcharge (Rs 3.23/kWh or Rs 0.43/kWh), Quarterly Tariff Adjustment (QTA), Fuel Price Adjustment (FPA), and 18% General Sales Tax (GST).
4. **Wholesale Purchase Price vs. Consumer-End Loss-Adjusted Component**: Under the 2026 Prosumer Regulations, press reports in February 2026 widely quoted a solar buyback rate of "Rs 10 to Rs 11 per unit" (or Rs 11.30/kWh), whereas NEPRA's formal statutory notifications (07-01-2026 CPPA-G PPP Decision, Para 36 and S.R.O. 1501(I)/2026 dated 07-09-2026, Para 23) explicitly establish the National Average Energy Purchase Price (NAEPP) at **Rs 8.13/kWh** (which translates to Rs 9.12/kWh at the consumer distribution level after accounting for 11.27% T&D losses).

Topic A: Residential (A-1) Slab Tariff, Single-Phase Meters, Sanctioned Load Under 5 kW

Complete Schedule of Residential Slab Tariffs (Category A-1(a))

For residential consumers with a sanctioned load below 5 kW connected via single-phase meters (Tariff Category A-1(a)), billing is governed by the Schedule of Tariff notified under S.R.O. 279(I)/2026 dated 12-02-2026 (implementing the NEPRA Tariff Rationalization Decision dated 11-02-2026), which amended the initial CY 2026 notification (S.R.O. 46(I)/2026 dated 13-01-2026) and superseded the FY 2025-26 notification (S.R.O. 1161(I)/2025 dated 01-07-2025). The table below provides the complete verification matrix for Topic A across primary and independent secondary sources (The Express Tribune, 14-02-2026, Business Recorder, 12-02-2026, Dawn, 12-02-2026, eBillPakistan, 2026, BijliBills, 2026, and SolarWorks, 2026).

Item / Consumer Slab	Verified Value (GoP Applicable vs. NEPRA Determined / Superseded)	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Lifeline: 1–50 Units (≤1 kW	**Rs 3.95/kWh** GoP Variable; **Nil (Rs 0)** Fixed Charge (Rs 75/mo min. charge).	12 Feb 2026 (S.R.O.	NEPRA SoT S.R.O.	Express Tribune;	High

Item / Consumer Slab	Verified Value (GoP Applicable vs. NEPRA Determined / Superseded)	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
load, ≤100 kWh 12-mo max)	*(NEPRA Determined CY26: Rs 36.13/kWh w/ PYA, Rs 35.43/kWh w/o PYA; FY25-26 NEPRA: Rs 6.76/kWh)*	279(I)/2026; rate unchanged from 01 Jul 2025)	1161/2025; CY26 Uniform Decision 12-01-2026	eBillPakistan; BijliBills	
Lifeline: 51–100 Units (≤1 kW load, ≤100 kWh 12-mo max)	**Rs 7.74/kWh** GoP Variable (whole-bill); **Nil (Rs 0)** Fixed Charge. *(NEPRA Determined CY26: Rs 36.13/kWh w/ PYA, Rs 35.37/kWh w/o PYA; FY25-26 NEPRA: Rs 11.50/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026; rate unchanged from 01 Jul 2025)	NEPRA SoT S.R.O. 1161/2025; CY26 Uniform Decision 12-01-2026	Express Tribune; eBillPakistan; BijliBills	High
Protected: 1–100 Units (<5 kW, ≤200 kWh/mo for 6 consecutive mos)	**Rs 10.54/kWh** GoP Variable; **Rs 200/kW/month** Fixed Charge. *(NEPRA Determined CY26: Rs 36.13/kWh w/ PYA; FY25-26 NEPRA: Rs 28.55/kWh; Fixed Charge was Nil before 12 Feb 2026)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	Express Tribune; Business Recorder; eBillPakistan	High
Protected: 101–200 Units (<5 kW, ≤200 kWh/mo for 6 consecutive mos)	**Rs 13.01/kWh** GoP Variable (incremental: first 100 @ Rs 10.54, 101–200 @ Rs 13.01); **Rs 300/kW/month** Fixed Charge. *(NEPRA Determined CY26: Rs 36.13/kWh; FY25-26: Rs 31.43/kWh; Fixed was Nil before 12 Feb 2026)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	Express Tribune; Business Recorder; BijliBills	High
Unprotected: 1–100 Units (<5 kW, Non-ToU)	**Rs 22.44/kWh** GoP Variable (whole-bill); **Rs 275/kW/month** Fixed Charge. *(NEPRA Determined CY26: Rs 36.13/kWh; FY25-26 NEPRA: Rs 30.27/kWh; Fixed was Nil before 12 Feb 2026)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	Express Tribune; eBillPakistan; SolarWorks	High
Unprotected: 101–200 Units (<5 kW, Non-ToU)	**Rs 28.91/kWh** GoP Variable (whole-bill); **Rs 300/kW/month** Fixed Charge. *(NEPRA Determined CY26: Rs 36.13/kWh; FY25-26 NEPRA: Rs 33.76/kWh; Fixed was Nil before 12 Feb 2026)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	Express Tribune; eBillPakistan; BijliBills	High
Unprotected: 201–300 Units (<5 kW, Non-ToU)	**Rs 33.10/kWh** GoP Variable (whole-bill); **Rs 350/kW/month** Fixed Charge. *(NEPRA Determined CY26: Rs 36.13/kWh; FY25-26 NEPRA: Rs 37.04/kWh; Fixed was Nil before 12 Feb 2026)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	Express Tribune; Business Recorder; eBillPakistan	High
Unprotected: 301–400 Units (<5 kW, Non-ToU)	**Rs 36.46/kWh** GoP Variable (whole-bill); **Rs 400/kW/month** Fixed Charge. *(Superseded Jul 2025/Jan 2026 GoP rate: Rs 37.99/kWh & Rs 200/conn/mo; NEPRA Determined CY26: Rs 35.58/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; Business Recorder; eBillPakistan	High

Item / Consumer Slab	Verified Value (GoP Applicable vs. NEPRA Determined / Superseded)	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Unprotected: 401–500 Units (<5 kW, Non-ToU)	**Rs 38.95/kWh** GoP Variable (whole-bill); **Rs 500/kW/month** Fixed Charge. *(Superseded Jul 2025/Jan 2026 GoP rate: Rs 40.20/kWh & Rs 400/conn/mo; NEPRA Determined CY26: Rs 35.23/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; Business Recorder; eBillPakistan	High
Unprotected: 501–600 Units (<5 kW, Non-ToU)	**Rs 40.22/kWh** GoP Variable (whole-bill); **Rs 675/kW/month** Fixed Charge. *(Superseded Jul 2025/Jan 2026 GoP rate: Rs 41.62/kWh & Rs 600/conn/mo; NEPRA Determined CY26: Rs 34.97/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; Business Recorder; eBillPakistan	High
Unprotected: 601–700 Units (<5 kW, Non-ToU)	**Rs 41.85/kWh** GoP Variable (whole-bill); **Rs 675/kW/month** Fixed Charge. *(Superseded Jul 2025/Jan 2026 GoP rate: Rs 42.76/kWh & Rs 800/conn/mo; NEPRA Determined CY26: Rs 34.81/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; Business Recorder; eBillPakistan	High
Unprotected: Above 700 Units (<5 kW, Non-ToU)	**Rs 47.20/kWh** GoP Variable (whole-bill); **Rs 675/kW/month** Fixed Charge. *(Superseded Jul 2025/Jan 2026 GoP rate: Rs 47.69/kWh & Rs 1,000/conn/mo; NEPRA Determined CY26: Rs 35.07/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; Business Recorder; eBillPakistan	High
Minimum Monthly Customer Charge (Where No Fixed Charge Applies)	**Rs 75/consumer/month** (Single-Phase); **Rs 150/consumer/month** (Three-Phase). Waived whenever monthly fixed charges are billed.	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-II, Clause 5)	LESCO Rebasing Decision 07-01-2026, p. 22	S.R.O. 1161/2025; eBillPakistan; BijliBills	High

Statutory Definitions, Category Transitions, and Slab Application Rules

The exact legal definitions of Lifeline, Protected, and Unprotected residential consumers, as well as the mechanics of how slabs are applied and how consumers transition between categories, are codified in Annex-V (*Terms and Conditions of Tariff (For Supply of Electric Power to Consumers by Distribution Licensees)*) of the NEPRA LESCO Tariff Determination dated 07-01-2026 (Pages 20–22) and Footnotes 1 through 4 of the Schedule of Tariff (S.R.O. 1161(I)/2025, Page 4):

1. **Lifeline Consumer Definition and Whole-Bill Application**: Under Part-I, Clause 9 and Part-II, Clause 2 of Annex-V (07-01-2026 LESCO Rebasing Decision, p. 20 & p. 22), a *"Lifeline Consumer"* is a residential Non-Time-of-Use consumer having a single-phase electric connection with a **sanctioned load up to 1 kW** whose maximum consumption across the **last twelve (12) months and the current billing month is less than or equal to 100 units (≤100 kWh)**. Two consumption bands exist: up to 50 units (Rs 3.95/kWh) and 51–100 units (Rs 7.74/kWh). Critically, Footnote 2 of the Schedule of Tariff explicitly states that under Lifeline Tariff, **"there will be no slab benefit."** Therefore, a Lifeline consumer using 45 kWh pays **45 × Rs 3.95 = Rs 177.75**, whereas a Lifeline consumer using 80 kWh pays **Rs 7.74/kWh on all 80 units** (**80 × Rs 7.74 = Rs 619.20**), not incremental billing across the 1–50 band. If a Lifeline consumer's monthly

consumption exceeds 100 kWh in any single month, the 12-month maximum exceeds 100 units and the consumer forfeits Lifeline status for the subsequent 12 months.

2. **Protected Consumer Definition and Incremental (Previous-Slab) Benefit**: Under Footnote 3 of the Schedule of Tariff (S.R.O. 1161(I)/2025, p. 4), to qualify as a Protected residential consumer, a connection must be a **Non-ToU residential consumer (sanctioned load under 5 kW)** consuming **≤200 kWh per month consistently for the past six (6) consecutive months**. Unlike Lifeline and Unprotected consumers, Footnote 3 stipulates that **"Residential Consumers will be given the benefit of one previous slab"** within the Protected category. Consequently, Protected consumers receive **incremental (telescopic) billing**:
   - A Protected consumer using **100 kWh or less** is billed at **Rs 10.54/kWh** on all units, plus the fixed charge for the 1–100 slab (**Rs 200/kW/month**).
   - A Protected consumer using between **101 and 200 kWh** (for example, 160 kWh) is billed **Rs 10.54/kWh on the first 100 units** and **Rs 13.01/kWh on the remaining 60 units** (**100 × 10.54 + 60 × 13.01 = Rs 1,834.60**), plus the fixed charge for the 101–200 slab (**Rs 300/kW/month**).
3. **Unprotected Consumer Definition and Whole-Bill (Landing-Slab) Application**: Any Non-ToU residential consumer (< 5 kW) who does not satisfy the Lifeline criteria or the 6-month consecutive ≤200 kWh Protected threshold falls into the **Unprotected** category. Under Footnote 4 of the Schedule of Tariff (S.R.O. 1161(I)/2025, p. 4), **"Under Un-Protected tariff there will be no previous slab benefit."** This rule — eliminated during the IMF structural reforms of 2023–2024 and maintained throughout 2025 and 2026—means that **whole-bill (landing-slab) pricing applies**: every single unit from unit 1 up to the final unit consumed in that billing cycle is charged at the single per-kWh rate of the highest slab reached. For example, an Unprotected consumer whose monthly meter reading shows 301 kWh does not pay tiered rates across 1–100, 101–200, and 201–300; instead, all 301 units are billed at the 301–400 slab rate of **Rs 36.46/kWh** (**301 × 36.46 = Rs 10,974.46**), whereas consuming 300 kWh costs **300 × 33.10 = Rs 9,930.00**—an immediate marginal cliff of **Rs 1,044.46** (plus a Rs 50/kW/month jump in fixed charges and taxes) for a single additional kilowatt-hour.
4. **Category Transition and 6-Month Rolling Window Mechanics**: Movement between Protected and Unprotected status is evaluated dynamically each month over a rolling 6-month look-back window (BijliBills, 2026; Dawn, 23-09-2026). If a Protected consumer consumes **201 kWh** in a summer month (say, June), the consumer immediately transitions to **Unprotected (201–300 slab at Rs 33.10/kWh + Rs 3.23/kWh FC surcharge)** for June. Moreover, even if the consumer reduces consumption back to 150 kWh in July, August, September, October, and November, the presence of the single 201 kWh month in June within the preceding 6-month history keeps the consumer in the **Unprotected 101–200 slab (Rs 28.91/kWh + Rs 3.23/kWh FC surcharge)** for six consecutive subsequent billing months until six full months of ≤200 kWh consumption have been re-established.
5. **Pro-Rata Consumption Adjustment Rule for Extended Reading Cycles**: Under Part-I, Clause 1 of Annex-V (07-01-2026 LESCO Rebasing Decision, p. 20) and Clause 6.1.1.1(a) of NEPRA's Consumer Service Manual (CSM Revised 2025), a billing month is defined as a period of 30 days. If LESCO records a meter reading beyond the number of days in that calendar month (e.g., 33 days in a 30-day month), LESCO is legally required to normalize the units on a pro-rata basis (**Charged Units = Actual Units × (Days in Calendar Month / Actual Reading Cycle Days)**) to determine slab landing and Protected status eligibility.

Fixed Charges Basis and Effective Date (S.R.O. 279(I)/2026)

Prior to 12 February 2026, residential fixed charges under S.R.O. 1161(I)/2025 applied solely to Unprotected consumers exceeding 300 units per month and were denominated on a flat **per-connection**

(Rs./Consumer/Month) basis (Rs 200 for 301–400, Rs 400 for 401–500, Rs 600 for 501–600, Rs 800 for 601–700, and Rs 1,000 above 700 units). Effective **12 February 2026**, under S.R.O. 279(I)/2026 and the NEPRA Tariff Rationalization Decision dated 11-02-2026, the basis of residential fixed charges was changed to **per kilowatt of sanctioned load per month (Rs./kW/Month)** and extended to all Protected and Unprotected slabs (excluding Lifeline consumers):

- **Protected 1–100 Units**: **Rs 200/kW/month**
- **Protected 101–200 Units**: **Rs 300/kW/month**
- **Unprotected 1–100 Units**: **Rs 275/kW/month**
- **Unprotected 101–200 Units**: **Rs 300/kW/month**
- **Unprotected 201–300 Units**: **Rs 350/kW/month**
- **Unprotected 301–400 Units**: **Rs 400/kW/month**
- **Unprotected 401–500 Units**: **Rs 500/kW/month**
- **Unprotected >500 Units (501–600, 601–700, >700)**: **Rs 675/kW/month**

Because most single-phase connections in Lahore have a sanctioned load of **1 kW to 4 kW** (with 1 kW or 2 kW being the standard entry on older domestic demand notices), a 2 kW single-phase Unprotected household consuming 350 kWh now pays **2 kW × Rs 400/kW = Rs 800/month** in fixed charges (plus 18% GST), compared to Rs 200/month under the pre-February 2026 regime (The Express Tribune, 14-02-2026). Under Part-II, Clause 5 of Annex-V (07-01-2026 LESCO Rebasing Decision, p. 22), the Minimum Monthly Customer Charge (Rs 75 single-phase / Rs 150 three-phase) applies **only** to consumers on whom no monthly fixed charges are levied (i.e., Lifeline consumers with zero consumption); for all consumers subject to fixed charges, no separate minimum monthly charge is levied even if zero energy is consumed.

Topic B: Time-of-Use (ToU) Residential Tariff (Category A-1(b))

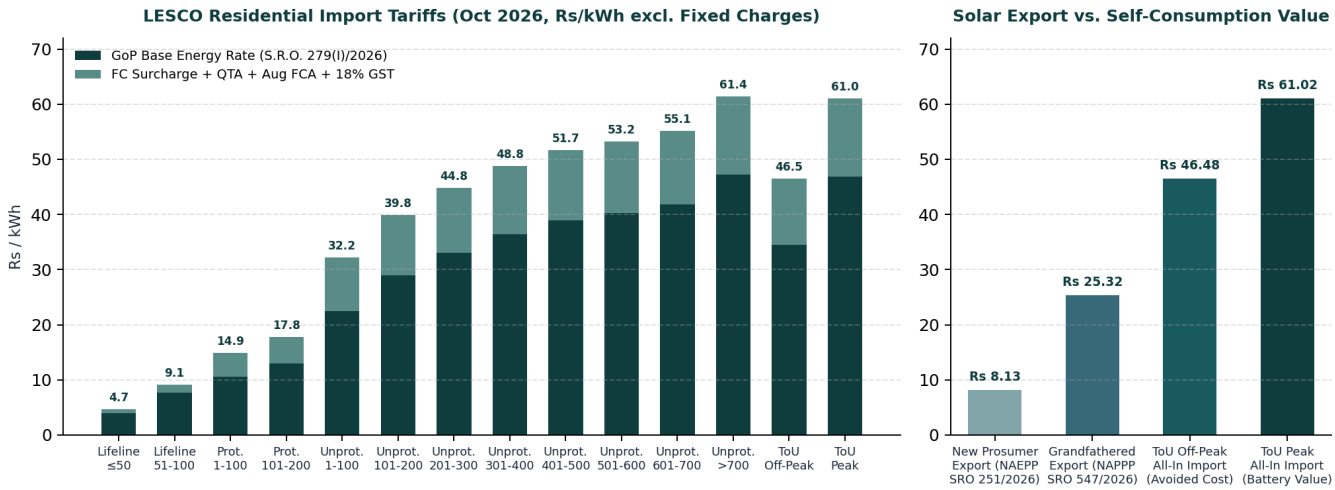


Figure 1: Verified LESCO Residential Base Variable vs. All-In Import Rates (including FC Surcharge, Q2 CY26 QTA, Aug 2026 FCA, and 18% GST) alongside Prosumer Export Credits (NAEPP Rs 8.13/kWh vs. Legacy NAPPP Rs 25.32/kWh) as of October 2026.

Peak and Off-Peak Rates, Eligibility, and Seasonal Windows

Residential connections with a sanctioned load of **5 kW and above** fall under Tariff Category **A-1(b) (Time-of-Use)**, while residential consumers with a new or existing Pre-Paid meter fall under Category **A-1(c) (Pre-Paid Supply Tariff)**. The table below presents the verified parameters for Topic B across primary NEPRA

determinations and independent secondary sources (The Express Tribune, 14-02-2026, Business Recorder, 12-02-2026, eBillPakistan, 2026, LESCO Online Bills, 2026, PakEra, 2026, and TheGreenOh, 2026).

Item / Parameter	Verified Value (GoP Applicable vs. NEPRA Determined / Superseded)	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Residential ToU Peak Energy Rate (A-1(b))	**Rs 46.85/kWh** GoP Applicable Variable Rate. *(NEPRA Determined CY26 Uniform: Rs 33.00/kWh w/ PYA, Rs 32.35/kWh w/o PYA; LESCO Standalone CY26: Rs 28.01/kWh; FY25-26 NEPRA: Rs 42.79/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026; unchanged from 01 Jul 2025 S.R.O. 1161/2025)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	Express Tribune; eBillPakistan; SolarWorks	High
Residential ToU Off-Peak Energy Rate (A-1(b))	**Rs 34.53/kWh** GoP Applicable Variable Rate (reduced by Rs 6.00/kWh from **Rs 40.53/kWh** in S.R.O. 1161/2025 & Jan 2026 SoT). *(NEPRA Determined CY26 Uniform: Rs 30.61/kWh w/ PYA, Rs 29.96/kWh w/o PYA; LESCO Standalone CY26: Rs 25.62/kWh)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; Business Recorder; eBillPakistan	High
Residential ToU Fixed Charge (A-1(b))	**Rs 675/kW/month** (billed on Sanctioned Load / Billing Demand; replaced flat **Rs 1,000/consumer/month** in S.R.O. 1161/2025). *(Discrepancy note: Annex-V Part-I Clause 4 defines Billing Demand as higher of 25% of sanctioned load or actual MDI; secondary guides cite 50% of sanctioned load or MDI)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; LESCO Rebasing Decision 07-01-2026, p. 20	Express Tribune; eBillPakistan; BijliBills	High
Pre-Paid Residential Supply Tariff (A-1(c))	**Rs 42.12/kWh** GoP Variable Rate (cut from **Rs 54.35/kWh** in S.R.O. 1161/2025); **Rs 675/kW/month** Fixed Charge. *(NEPRA Determined CY26 Uniform: Rs 34.47/kWh w/ PYA)*	12 Feb 2026 (S.R.O. 279(I)/2026 & NEPRA Decision 11-02-2026)	NEPRA Decision 11-02-2026; CY26 Uniform Decision 12-01-2026	S.R.O. 1161/2025; Express Tribune; eBillPakistan	High
ToU Eligibility & Mandatory Threshold	**Mandatory** for sanctioned load **≥ 5 kW** (Three-Phase 400V ToU meter required). **Optional** for sanctioned load **< 5 kW** upon consumer request.	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-I Clause 11 & Part-II Clauses 3–4)	LESCO Rebasing Decision 07-01-2026, pp. 21–22; S.R.O. 1161/2025, p. 4	eBillPakistan; BijliBills; SolarWorks	High
Peak Hours: Winter (Dec to Feb)	**5:00 PM to 9:00 PM** (17:00–21:00 PKT, 4 hours/day); Off-Peak = remaining 20 hours.	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-I	LESCO Rebasing Decision 07-01-2026, p. 21	LESCO Online Bills; PakEra; TheGreenOh	High

Item / Parameter	Verified Value (GoP Applicable vs. NEPRA Determined / Superseded)	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
		Clause 10, p. 21)			
Peak Hours: Spring (Mar to May)	**6:00 PM to 10:00 PM** (18:00–22:00 PKT, 4 hours/day); Off-Peak = remaining 20 hours.	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-I Clause 10, p. 21)	LESCO Rebasing Decision 07-01-2026, p. 21	LESCO Online Bills; PakEra; TheGreenOh	High
Peak Hours: Summer (Jun to Aug)	**7:00 PM to 11:00 PM** (19:00–23:00 PKT, 4 hours/day); Off-Peak = remaining 20 hours.	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-I Clause 10, p. 21)	LESCO Rebasing Decision 07-01-2026, p. 21	LESCO Online Bills; PakEra; TheGreenOh	High
Peak Hours: Autumn (Sep to Nov)	**6:00 PM to 10:00 PM** (18:00–22:00 PKT, 4 hours/day); Off-Peak = remaining 20 hours.	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-I Clause 10, p. 21)	LESCO Rebasing Decision 07-01-2026, p. 21	LESCO Online Bills; PakEra; TheGreenOh	High
Residential vs. Commercial/Industrial Peak Hours	**Identical** across all LESCO ToU consumer categories (Residential A-1(b), Commercial A-2(c), Industrial B-1(b)/B-2(b)/B-3/B-4, Bulk C, Agricultural D).	07 Jan 2026 (LESCO Rebasing Decision Annex-V, Part-I Clause 10, p. 21)	LESCO Rebasing Decision 07-01-2026, p. 21	LESCO Online Bills; PakEra; TheGreenOh	High

Analysis of ToU Spread, Eligibility Mechanics, and Billing Demand Basis

Under the 12 February 2026 tariff rationalization (S.R.O. 279(I)/2026), the Government of Pakistan maintained the residential ToU **Peak energy charge at Rs 46.85/kWh** while reducing the **Off-Peak energy charge by Rs 6.00/kWh from Rs 40.53/kWh to Rs 34.53/kWh** (The Express Tribune, 14-02-2026; eBillPakistan, 2026). For battery energy-management systems in Lahore, this revision nearly doubled the base Peak-to-Off-Peak price spread from **Rs 6.32/kWh** (under the July 2025 – January 2026 schedule) to **Rs 12.32/kWh** before taxes, or **Rs 14.54/kWh** after applying 18% GST (**12.32 × 1.18**). When comparing solar self-consumption or battery discharge during the 4-hour evening Peak window (saving Rs 46.85/kWh base + Rs 3.23/kWh FC + Rs 0.5194/kWh QTA + Rs 1.1086/kWh FCA + 18% GST ≈ **Rs 60.83/kWh** all-in) against exporting daytime solar to the grid under the 2026 Prosumer Net-Billing regime (credited at **Rs 8.13/kWh** NAEPP with zero tax credit), evening peak-shaving delivers a **7.48× higher economic return per kilowatt-hour** than daytime grid export.

With respect to eligibility and connection phase rules, Annex-V of the LESCO CY 2026 Tariff Determination dated 07-01-2026 (Part-I Clause 11 and Part-II Clauses 3 & 4, pp. 21–22) establishes three binding rules:

1. **Sanctioned Load ≥ 5 kW (Mandatory 3-Phase ToU)**: All consumers having a sanctioned load of **5 kW and above** shall be provided a **3-Phase 400 Volts** supply and billed under the Time-of-Use (ToU) tariff (Category A-1(b)). Under Part-II, Clause 3, existing residential consumers with sanctioned load of 5 kW and above who were historically provided single-phase supply must be converted to three-phase supply by the distribution company.
2. **Sanctioned Load < 5 kW (Optional ToU)**: Under the statutory Note in the Schedule of Tariff (S.R.O. 1161(I)/2025, p. 4), *"Residential consumers having sanctioned load less than 5 kW can opt for ToU metering and shall be billed under ToU tariff."* However, once billed under ToU, they lose slab benefits and pay the Rs 675/kW/month fixed charge.
3. **Billing Demand Discrepancy for Fixed Charges**: Under S.R.O. 279(I)/2026, the ToU fixed charge is **Rs 675/kW/month**. A notable discrepancy exists between the general statutory definition in NEPRA's *Terms and Conditions of Tariff* and secondary billing guides regarding how the billing kilowatts are determined:
    - **Primary Statutory Definition (25% Floor)**: Part-I, Clause 4 of Annex-V (07-01-2026 LESCO Rebasing Decision, p. 20) defines *"Billing Demand"* as *"the **25% of the sanction load or Actual maximum demand recorded in a month, whichever is higher**, except in the case of agriculture tariff D2 where 'Billing Demand' shall mean the sanctioned load."*
    - **Secondary Billing Portals & Fixed Charge Practice (50% Floor vs. Full Sanctioned Load)**: Secondary tariff guides (eBillPakistan, 2026; BijliBills, 2026) state that for residential ToU connections (≥ 5 kW), fixed charges are calculated on **50% of sanctioned load or actual Maximum Demand Indicator (MDI) recorded during the month, whichever is higher** (mirroring the historical industrial/commercial 50% MDI floor prior to the June 2024 reduction to 25%), whereas for non-MDI or standard domestic connections LESCO's billing system often multiplies Rs 675 directly by the integer **Sanctioned Load (kW)** printed on the bill unless MDI exceeds sanctioned load. For EMS configuration, software should allow user selection between `100% Sanctioned Load`, `max(50% Sanctioned Load, MDI)`, and `max(25% Sanctioned Load, MDI)`.

Finally, regarding whether residential ToU peak hours differ from commercial or industrial peak hours: **they do not differ**. Part-I, Clause 10 of Annex-V (07-01-2026 LESCO Rebasing Decision, p. 21) defines a single, uniform table of Peak and Off-Peak hours for the entire LESCO service territory across all ToU categories (Residential A-1(b), Commercial A-2(c), General Services A-3(b), Industrial B-1(b)/B-2(b)/B-3/B-4, Single Point Supply C, and Agricultural D-1(b)/D-2(b)):

- **December to February (inclusive)**: **5:00 PM to 9:00 PM** (Peak); remaining 20 hours (Off-Peak).
- **March to May (inclusive)**: **6:00 PM to 10:00 PM** (Peak); remaining 20 hours (Off-Peak).
- **June to August (inclusive)**: **7:00 PM to 11:00 PM** (Peak); remaining 20 hours (Off-Peak).
- **September to November (inclusive)**: **6:00 PM to 10:00 PM** (Peak); remaining 20 hours (Off-Peak).

Topic C: Bill Components, Surcharges, Taxes, and Worked Bill Calculations

Verification of All LESCO Residential Bill Components (CY 2026)

Beyond base variable energy charges and monthly fixed charges, a LESCO residential electricity bill incorporates pass-through fuel and quarterly tariff adjustments, statutory debt-servicing surcharges, indirect taxes, and conditional withholding taxes. The table below synthesizes the verified values, applicability rules, and primary/secondary sources (NEPRA August 2026 FCA Decision & S.R.O. 1754(I)/2026 dated 07-10-2026,

S.R.O. 1501(I)/2026 dated 07-09-2026, LESCO Rebasing Decision 07-01-2026, Dawn, Business Recorder, BillAsaan, Qaaf, Oraan, and LESCO Online Bills).

Bill Component	Verified Value & Recent 2026 Trajectory	Consumer Category Applicability	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Fuel Price Adjustment (FPA / FCA)	**+Rs 1.1086/kWh** in **Oct 2026 bills** (for Aug 2026 units, S.R.O. 1754(I)/2026). Prior months: **+Rs 2.0581/kWh** in Sep 2026 (for Jul 2026, S.R.O. 1499(I)/2026); **+Rs 0.7503/kWh** in Aug 2026 (for Jun 2026); **−Rs 0.1502/kWh** in Jul 2026 (for May 2026). Requested for Nov 2026 (Sep 2026 units): **+Rs 2.5624/kWh**.	Applies to **Protected, Unprotected, and ToU** residential consumers (billed 2 months in arrears on reference month's kWh). **Exempt**: Lifeline (≤100 kWh, ≤1 kW) and Prepaid (A-1(c)) consumers.	07 Oct 2026 (S.R.O. 1754(I)/2026 for Oct 2026 billing)	NEPRA FCA Decision & S.R.O. 1754(I)/2026 dated 07-10-2026; LESCO Tariff Index	Business Recorder; Qaaf; BillAsaan	High
Quarterly Tariff Adjustment (QTA)	**+Rs 0.5194/kWh** for **Sep–Nov 2026** (Q2 CY26 QTA, S.R.O. 1501(I)/2026). Prior quarters: **−Rs 1.9857/kWh** rebate for Jun–Aug 2026 (Q1 CY26, S.R.O. 953(I)/2026); **+Rs 0.3504/kWh** for Mar–May 2026 (Q2 FY26, S.R.O. 459(I)/2026).	Applies to **Protected, Unprotected, and ToU** residential consumers on current month's imported kWh. **Exempt**: Lifeline and Prepaid consumers.	07 Sep 2026 (S.R.O. 1501(I)/2026 for Sep, Oct & Nov 2026)	S.R.O. 1501(I)/2026 dated 07-09-2026; LESCO Tariff Index	Dawn; Business Recorder; eBillPakistan	High
Financing Cost (FC) Surcharge / DSS (PHL)	**Rs 3.23/kWh** for Unprotected & ToU consumers; **Rs 0.43/kWh** for Protected (≤200 units) consumers; **Rs 0 (Exempt)** for Lifeline consumers.	Billed per kWh imported on current month's units to service Power Holding Limited (PHL) / Rs 1.225T circular debt facility.	Ongoing in CY 2026 (re-affirmed under Jun 2025 / CY 2026 debt servicing notifications)	NEPRA LESCO Tariff Notifications	eBillPakistan; BijliBills; LESCO Online Bills	High
Electricity Duty (ED — Provincial)	**Discrepancy**: Statutorily announced as **0% (Discontinued via bills)** effective **01 Jul 2025**; however, **1.5%** of (Variable Energy + QTA + FPA) continues to appear on many actual 2026 LESCO bills.	Historically applied to all non-Lifeline residential consumers at **1.5%** of variable energy charges (and 1.5% on FPA).	01 Jul 2025 discontinuation announcement vs. ongoing 2026 LESCO billing practice	Power Division / Provincial Budget Directives 2025–2026	LESCO Online Bills; Qaaf; Oraan	High (Discrepancy Documented)
General Sales Tax (GST —	**18%** standard rate on taxable bill value (Variable Energy + Fixed Charges +	Applies to **all residential categories**	Effective since Feb 2023	LESCO Rebasing Decision 07-	BillAsaan; Oraan; LESCO	High

Bill Component	Verified Value & Recent 2026 Trajectory	Consumer Category Applicability	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Section 3(1) Sales Tax Act)	QTA + FC Surcharge + ED + Meter Rent, plus 18% on FPA + ED on FPA).	(Lifeline, Protected, Unprotected, ToU, Prepaid).	Finance (Supplementary) Act; active throughout CY 2026	01-2026, Annex-V Part-I Clause 15	Online Bills	
TV License Fee (PTV Fee)	**Rs 0 (Abolished)**. *(Superseded value: Rs 35/month per domestic meter prior to 01 Jul 2025).*	Abolished across **all residential and commercial** consumers nationwide.	01 Jul 2025	Federal Budget FY 2025-26 / Power Division Directive	LESCO Online Bills; BillAsaan; BijliBills	High
Advance / Withholding Income Tax (Sec. 235 ITO 2001)	**0% (Nil)** for Active Taxpayers List (ATL / Filers) regardless of bill amount; **0% (Nil)** for Non-Filers if monthly bill **< Rs 25,000**; **7.5%** of total bill for **Non-Filers** if monthly bill **≥ Rs 25,000**.	Domestic (A-1) consumers. (Note: Extra Tax / Further Tax apply strictly to unregistered commercial/industrial connections, not domestic A-1).	Active throughout CY 2026 (Finance Act provisions under Section 235)	Income Tax Ordinance 2001, Section 235 / Division IV Part IV First Schedule	BillAsaan; Oraan; LESCO Online Bills	High

Detailed Mechanics and Statutory Basis of Each Bill Component

1. **Fuel Price Adjustment (FPA / FCA)**: Governed by Section 31(7) of the NEPRA Act and Annex-I of the LESCO CY 2026 Tariff Determination dated 07-01-2026 (Pages 13–14), the monthly Fuel Charges Adjustment measures the variance between actual generation fuel cost per kWh and the monthly reference fuel cost established in Annex-IV of the determination. Critically, FPA is **always billed with a two-month lag** based on the **units consumed in that earlier reference month**, not the current month's units. For instance, in **October 2026 bills**, LESCO applies the **August 2026 FCA of +Rs 1.1086/kWh** (approved via NEPRA Decision No. D-20276-94 & S.R.O. 1754(I)/2026 dated 07-10-2026) multiplied by the consumer's **August 2026 kWh consumption**. On the LESCO bill, FPA attracts its own 18% GST (and 1.5% ED where levied), printed in the adjustment box. Lifeline consumers (≤100 units, ≤1 kW) and Prepaid consumers are exempt from both positive and negative FPA adjustments.
2. **Quarterly Tariff Adjustment (QTA)**: Governed by Annex-I-A of the LESCO CY 2026 Tariff Determination dated 07-01-2026 (Pages 15–16), QTA reconciles quarterly variations in Capacity Purchase Price (CPP), Variable O&M, Use of System Charges (UoSC), Market Operator Fee (MOF), and the impact of T&D losses on monthly FCA. Unlike FPA, QTA is applied to the **current billing month's imported kWh** over a three-month recovery window. For the three-month period of **September, October, and November 2026**, the applicable QTA notified via S.R.O. 1501(I)/2026 dated 07-09-2026 is **+Rs 0.5194/kWh** (replacing the Q1 CY 2026 negative QTA rebate of **−Rs 1.9857/kWh** that applied during June, July, and August 2026 under S.R.O. 953(I)/2026). Lifeline and Prepaid consumers are exempt from positive QTA charges (though NEPRA passes negative QTA relief to Protected and Unprotected consumers alike).
3. **Financing Cost (FC) Surcharge / Debt Servicing Surcharge (DSS)**: Instituted by the Federal Government under Section 31 of the NEPRA Act to service circular debt obligations parked in Power Holding Limited (PHL) and the June 2025 Rs 1.225 trillion commercial bank refinancing package, the FC

Surcharge is levied on current-month imported units at **Rs 3.23/kWh** for Unprotected and ToU residential consumers and **Rs 0.43/kWh** for Protected residential consumers (≤200 units), while Lifeline consumers are exempt (eBillPakistan, 2026; BijliBills, 2026).

4. **Electricity Duty (ED) Discrepancy**: In June 2025, the Federal Ministry of Energy wrote to provincial Chief Ministers and announced the discontinuation of provincial Electricity Duty collection through DISCO bills starting **1 July 2025** (LESCO Online Bills, 2026). However, empirical inspection of 2026 LESCO bills and billing analyses (Qaaf, 2026; Oraan, 2026) confirms that **1.5% Electricity Duty** on variable energy charges (and on FPA) has continued to appear on many LESCO residential bills in Lahore where provincial finance statutes remained active. In the worked examples below, both the **Without ED (0%)** and **With ED (1.5%)** totals are explicitly shown so that EMS financial models can toggle the flag accurately.
5. **General Sales Tax (GST), PTV Fee, and Section 235 Withholding Income Tax**:
   - **GST (18%)**: Charged at **18%** under the Sales Tax Act, 1990 on the sum of Cost of Electricity (Variable Energy Charges), Fixed Charges, QTA, FC Surcharge, Electricity Duty (if levied), and Meter Rent, plus 18% on FPA (and ED on FPA).
   - **TV License Fee (PTV Fee)**: **Rs 0**. The historical Rs 35/month PTV fee was abolished across Pakistan effective **1 July 2025** (BillAsaan, 2026; LESCO Online Bills, 2026).
   - **Income / Withholding Tax (Section 235 of Income Tax Ordinance, 2001)**: Domestic consumers appearing on FBR's Active Taxpayers List (**Filers / ATL**) pay **0% (Nil)** income tax on domestic electricity bills regardless of bill size. **Non-Filers** pay **0% (Nil)** if the total monthly bill is **less than Rs 25,000**, and **7.5%** of the total bill amount if the monthly bill is **Rs 25,000 or above** (BillAsaan, 2026; Oraan, 2026).

Worked Example 1: Single-Phase Slab Consumer (Unprotected A-1(a)) — October 2026 Billing Cycle

Consider a standard single-phase Unprotected residential household in Lahore (Category A-1(a)) with the following verified parameters for the **October 2026** billing month (sources: S.R.O. 279(I)/2026, S.R.O. 1501(I)/2026, S.R.O. 1754(I)/2026, eBillPakistan):

- **Sanctioned Load**: **L = 2 kW** (Single-Phase)
- **Current Month Consumption (Oct 2026)**: E_{Oct} **= 350 kWh** (lands in **Unprotected 301–400 Units** slab; whole-bill rate R_{slab} **= Rs 36.46/kWh**; Fixed Charge rate F_{rate} **= Rs 400/kW/month**)
- **Reference Month Consumption for FPA (Aug 2026)**: E_{Aug} **= 400 kWh** (August 2026 FCA rate R_{FCA} **= +Rs 1.1086/kWh** per S.R.O. 1754(I)/2026)

The general LESCO residential slab bill formula is:

$$\text{Total Bill} = (E_{Oct} \times R_{slab})_{\text{Variable Energy}} + (L \times F_{rate})_{\text{Fixed Charge}} + (E_{Oct} \times R_{QTA})_{QTA} + (E_{Oct} \times R_{FC})_{\text{FC Surcharge}} + ED_{cur} + GST_{cur} + (E_{Aug} \times R_{FCA} + ED_{FCA} + GST_{FCA})_{\text{Gross FPA Block}} + IT_{Sec235}$$

Line Item	Formula & Inputs	Amount (Without 1.5% ED)	Amount (With 1.5% ED)	Governing Source
1. Cost of Electricity (Variable Energy)	**350 kWh × Rs 36.46/kWh** (whole-bill, no previous slab benefit)	**Rs 12,761.00**	**Rs 12,761.00**	S.R.O. 279(I)/2026

Line Item	Formula & Inputs	Amount (Without 1.5% ED)	Amount (With 1.5% ED)	Governing Source
2. Monthly Fixed Charges	**2 kW × Rs 400/kW/month** (301–400 slab)	**Rs 800.00**	**Rs 800.00**	S.R.O. 279(I)/2026
3. Quarterly Tariff Adjustment (QTA)	**350 kWh × Rs 0.5194/kWh** (Q2 CY26 QTA for Sep–Nov 2026)	**Rs 181.79**	**Rs 181.79**	S.R.O. 1501(I)/2026
4. Financing Cost (FC) Surcharge	**350 kWh × Rs 3.23/kWh** (Unprotected rate)	**Rs 1,130.50**	**Rs 1,130.50**	eBillPakistan
5. Electricity Duty (Current Month)	**0%** vs. **1.5% × (Rs 12,761.00 + Rs 181.79)**	**Rs 0.00**	**Rs 194.14**	Oraan / LESCO Online Bills
6. Current Pre-GST Subtotal	Sum of Lines 1 to 5	**Rs 14,873.29**	**Rs 15,067.43**	LESCO Bill Formula
7. General Sales Tax (18% on Current)	**18% × Line 6**	**Rs 2,677.19**	**Rs 2,712.14**	Sales Tax Act Sec. 3(1)
8. Fuel Price Adjustment (Aug 2026 Base)	**400 kWh (Aug) × Rs 1.1086/kWh**	**Rs 443.44**	**Rs 443.44**	S.R.O. 1754(I)/2026
9. ED on FPA + 18% GST on FPA	Without ED: **18% × 443.44 = 79.82**; With ED: **6.65 ED + 18% × (443.44+6.65) = 81.02**	**Rs 79.82**	**Rs 87.67**	Qaaf
10. TV License Fee & Income Tax (Sec. 235)	PTV = **Rs 0** (abolished); Income Tax = **Rs 0** (Bill < Rs 25,000 for both Filer & Non-Filer)	**Rs 0.00**	**Rs 0.00**	BillAsaan
TOTAL PAYABLE WITHIN DUE DATE	Sum of all components (rounded to nearest rupee: **Rs 18,074** vs. **Rs 18,311**)	**Rs 18,073.74**	**Rs 18,310.68**	**Effective All-In: Rs 51.64–52.32/kWh**

Worked Example 2: Three-Phase Time-of-Use Consumer (Category A-1(b)) — October 2026 Billing Cycle

Consider a three-phase residential ToU household in Lahore (Category A-1(b)) with the following parameters for **October 2026** (sources: S.R.O. 279(I)/2026, S.R.O. 1501(I)/2026, S.R.O. 1754(I)/2026, The Express Tribune):

- **Sanctioned Load**: **L = 5 kW** (Three-Phase 400V ToU Meter; Fixed Charge rate F_{rate} = **Rs 675/kW/month**)
- **Current Month Consumption (Oct 2026)**: Total E_{Oct} = **500 kWh**, split into **Peak (6:00 PM–10:00 PM)** E_{peak} = **120 kWh** at R_{peak} = **Rs 46.85/kWh** and **Off-Peak** E_{off} = **380 kWh** at R_{off} = **Rs 34.53/kWh**
- **Reference Month Consumption for FPA (Aug 2026)**: E_{Aug} = **600 kWh** (August 2026 FCA rate R_{FCA} = **+Rs 1.1086/kWh**)

The general LESCO residential ToU bill formula is:

$$\text{Total Bill} = (E_{peak} \times R_{peak} + E_{off} \times R_{off})_{\text{ToU Variable Energy}} + (L \times F_{rate})_{\text{Fixed Charge}} + E_{Oct} \times (R_{QTA} + R_{FC})_{\text{QTA + FC Surcharge}} + ED_{cur} + GST_{cur} + \text{Gross FPA Block} + IT_{Sec235}$$

Line Item	Formula & Inputs	Amount (Without 1.5% ED)	Amount (With 1.5% ED)	Governing Source
1a. Peak Variable Energy Charges	**120 kWh × Rs 46.85/kWh** (Oct Peak window: 6:00 PM – 10:00 PM)	**Rs 5,622.00**	**Rs 5,622.00**	S.R.O. 279(I)/2026
1b. Off-Peak Variable Energy Charges	**380 kWh × Rs 34.53/kWh** (Remaining 20 hours/day)	**Rs 13,121.40**	**Rs 13,121.40**	S.R.O. 279(I)/2026
1. Total Variable Energy Charges	**Rs 5,622.00 + Rs 13,121.40** (500 kWh total)	**Rs 18,743.40**	**Rs 18,743.40**	S.R.O. 279(I)/2026
2. Monthly Fixed Charges (5 kW)	**5 kW × Rs 675/kW/month** *(Note: if billed at 50% MDI floor = Rs 1,687.50; if 25% MDI floor = Rs 843.75)*	**Rs 3,375.00**	**Rs 3,375.00**	S.R.O. 279(I)/2026
3. Quarterly Tariff Adjustment (QTA)	**500 kWh × Rs 0.5194/kWh** (Q2 CY26 QTA for Sep–Nov 2026)	**Rs 259.70**	**Rs 259.70**	S.R.O. 1501(I)/2026
4. Financing Cost (FC) Surcharge	**500 kWh × Rs 3.23/kWh** (ToU / Unprotected rate)	**Rs 1,615.00**	**Rs 1,615.00**	eBillPakistan
5. Electricity Duty (Current Month)	**0%** vs. **1.5% × (Rs 18,743.40 + Rs 259.70)**	**Rs 0.00**	**Rs 285.05**	Oraan
6. Current Pre-GST Subtotal	Sum of Lines 1 to 5	**Rs 23,993.10**	**Rs 24,278.15**	LESCO Bill Formula
7. General Sales Tax (18% on Current)	**18% × Line 6**	**Rs 4,318.76**	**Rs 4,370.07**	Sales Tax Act Sec. 3(1)
8. Fuel Price Adjustment (Aug 2026 Base)	**600 kWh (Aug) × Rs 1.1086/kWh**	**Rs 665.16**	**Rs 665.16**	S.R.O. 1754(I)/2026
9. ED on FPA + 18% GST on FPA	Without ED: **18% × 665.16 = 119.73**; With ED: **9.98 ED + 18% × (665.16+9.98) = 121.53**	**Rs 119.73**	**Rs 131.51**	Qaaf
10. Subtotal Before Income Tax	Sum of Lines 6 to 9 (exceeds Rs 25,000 threshold!)	**Rs 29,096.75**	**Rs 29,444.89**	LESCO Bill Formula
11a. Total Payable — ATL Filer (0% IT)	0% Advance Income Tax for Active Taxpayer List (ATL) Filer	**Rs 29,096.75**	**Rs 29,444.89**	Sec. 235 ITO 2001
11b. Total Payable — Non-Filer (+7.5% IT)	**+7.5% Sec. 235 Tax** (**Rs 2,182.26** vs. **Rs 2,208.37** added because bill ≥ **Rs 25,000**)	**Rs 31,279.01**	**Rs 31,653.26**	Sec. 235 ITO 2001

Topic D: Solar and Net-Metering / Net-Billing Rules (NEPRA Prosumer Regulations 2026)

Complete Verification Matrix for Solar and Prosumer Rules (CY 2026)

The regulatory landscape for residential solar in Lahore shifted fundamentally on **9 February 2026** when NEPRA notified the NEPRA (Prosumer) Regulations, 2026 via S.R.O. 251(I)/2026, repealing the NEPRA (Alternative & Renewable Energy) Distributed Generation and Net Metering Regulations, 2015 (S.R.O. 892(I)/2015). Following widespread consumer backlash, parliamentary intervention by Prime Minister

Shehbaz Sharif, and writ petitions before the Lahore and Islamabad High Courts, NEPRA enacted two major amendments: S.R.O. 547(I)/2026 dated 02-04-2026 and S.R.O. 1330(I)/2026 dated 06-08-2026. The table below summarizes the verified framework across primary NEPRA statutory texts and independent secondary sources (Dawn, 10-02-2026, Dawn, 15-02-2026, Dawn, 03-04-2026, Business Recorder, 10-02-2026, Business Recorder, 28-08-2026, The Express Tribune, 03-04-2026, and LexMoor / RIAA Barker Gillette, 2026).

Regulatory Parameter	Verified Value & Legal Mechanism	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Principal Framework & Repeal	**NEPRA (Prosumer) Regulations, 2026 — S.R.O. 251(I)/2026**. Repeals the 2015 Net Metering Regulations (S.R.O. 892(I)/2015) under Reg. 21(1) and replaces 1:1 kWh net metering with **Net Billing** (gross import billing vs. NAEPP export credit) for new prosumers.	09 Feb 2026 (at once upon Gazette publication)	S.R.O. 251(I)/2026 dated 09-02-2026	Dawn (10-02-2026); Business Recorder (10-02-2026); RIAA Barker Gillette	High
First Amendment (Grandfathering Restoration)	**S.R.O. 547(I)/2026**: Substitutes Reg. 21(2), restoring 1:1 unit netting and **Rs 25.32/kWh** off-peak/NAPPP pricing for all valid agreements executed **before 9 Feb 2026** until expiry of their existing 7-year term (retroactively effective from 9 Feb 2026).	02 Apr 2026 (retroactive to **09 Feb 2026**)	S.R.O. 547(I)/2026 dated 02-04-2026	Dawn (03-04-2026); Express Tribune (03-04-2026); Business Recorder	High
Second Amendment (≤25 kW Concurrence Exemption)	**S.R.O. 1330(I)/2026** *(also cited in press as S.R.O. 1320(I)/2026)*: Amends Regs. 3 & 4 to exempt distributed generation facilities of **25 kW or below** from seeking NEPRA concurrence (saving Rs 1,000/kW fee and delegating approval to DISCO/LESCO).	06 Aug 2026	S.R.O. 1330(I)/2026 dated 06-08-2026	Business Recorder (28-08-2026); Dawn; Express Tribune	High
Export / Buyback Rate — New Prosumers (≥9 Feb 2026)	**Rs 8.13/kWh** (**National Average Energy Purchase Price — NAEPP**). *(Discrepancy: Feb 2026 press reports quoted **Rs 10–11/kWh** or Rs 11.30/kWh; consumer-end loss-adjusted energy component is **Rs 9.12/kWh**, while official CPPA-G/NEPRA NAEPP is strictly **Rs 8.13/kWh**).*	09 Feb 2026 (NAEPP value set on 07 Jan 2026 & confirmed 07 Sep 2026)	CPPA-G PPP Decision 07-01-2026, Para 36, p. 17; S.R.O. 1501(I)/2026, Para 23	Dawn (10-02-2026); Business Recorder (10-02-2026); Express Tribune (03-04-2026)	High (Discrepancy Explained)
Export / Buyback Rate — Legacy Grandfathered (<9 Feb 2026)	**1:1 kWh Netting** against imported Off-Peak units monthly + **Rs 25.32/kWh** (**National Average Power Purchase Price — NAPPP**) for net quarterly excess kWh. *(FY25-26 rate was ~Rs 25.98/kWh; prior to Dec 2024 was Rs 27/kWh).*	07 Jan 2026 (NAPPP value) & 02 Apr 2026 (S.R.O. 547(I)/2026)	CPPA-G PPP Decision 07-01-2026, Para 36, p. 17; S.R.O. 1501(I)/2026, Para 23	Dawn (03-04-2026); Express Tribune (03-04-2026); Business Recorder	High
Contract Duration & Renewal Rules	**New Agreements**: **5 years** from connection date (renewable by mutual consent for 5-year terms	09 Feb 2026 & 02 Apr 2026	S.R.O. 251(I)/2026 Reg. 9 & Reg.	Dawn (03-04-2026); Express	High

Regulatory Parameter	Verified Value & Legal Mechanism	Effective Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
	under Reg. 9). **Legacy Agreements**: Original **7-year** term honored; upon expiry/renewal, transitions to 5-year Net-Billing regime under Reg. 21(3). Any material modification changing max output triggers immediate transition to 2026 rules.		21; S.R.O. 547(I)/2026	Tribune (03-04-2026); RIAA Barker Gillette	
Phase & Tariff Category Eligibility	**Three-Phase (400V or 11kV) Only** under Reg. 2(1)(v) and Schedule-I. Single-phase (<5 kW slab) consumers cannot get net metering/net billing without first upgrading to a 3-phase connection (which places ≥5 kW consumers onto ToU tariff A-1(b)).	09 Feb 2026	S.R.O. 251(I)/2026 Reg. 2(1)(v) & Schedule-I	SolarWorks; eBillPakistan; RIAA Barker Gillette	High
System Size Limits & Transformer Cap	**100% of Sanctioned Load** (Reg. 3(2); reduced from 150% / 1.5× under 2015 rules), within overall range of **1 kW to 1 MW**. **80% Transformer Hosting Cap**: LESCO shall not entertain applications if DG on a distribution transformer reaches **80% of its rated capacity** (Reg. 3(5)). Load flow study required for **≥250 kW** (Reg. 3(3)).	09 Feb 2026	S.R.O. 251(I)/2026 Reg. 3(2), 3(3) & 3(5)	Dawn (10-02-2026); Business Recorder (10-02-2026); RIAA Barker Gillette	High

Deep-Dive Analysis of Net-Billing Mechanics, Buyback Discrepancy, and Grandfathering

1. How the Export / Buyback Rate is Calculated and Why Sources Cite Rs 8.13 vs. Rs 10–11/kWh

Under Regulation 14(1) of the NEPRA (Prosumer) Regulations, 2026 (S.R.O. 251(I)/2026, Page 7):

- The distribution licensee (LESCO) bills the prosumer at the end of each billing cycle strictly for the **gross electricity supplied (imported) by the licensee** to the prosumer at the applicable consumer-end tariff (e.g., Rs 46.85/kWh Peak and Rs 34.53/kWh Off-Peak, plus Rs 3.23/kWh FC Surcharge, Rs 0.5194/kWh QTA, FPA, and 18% GST).
- Simultaneously, the licensee raises an invoice/credit for the **electricity supplied (exported) by the prosumer to the licensee** at the **National Average Energy Purchase Price (NAEPP)**.
- Under Regulation 14(2), if the monetary value of the prosumer's exported kWh exceeds the monetary bill for imported electricity, the **net rupee amount is credited against the prosumer's next monthly billing cycle or paid by the licensee to the prosumer quarterly**. Crucially, **1:1 kWh unit offsetting is abolished for new prosumers**: daytime exported solar kWh can no longer cancel out nighttime off-peak imported kWh on a unit-for-unit basis.

Why do primary NEPRA documents state **Rs 8.13/kWh** while major newspapers (Dawn, 10-02-2026; Business Recorder, 10-02-2026; The Express Tribune, 03-04-2026) reported **"Rs 10 to Rs 11 per unit"** (or Rs 11.30/kWh)? The exact mathematical breakdown is documented in Paragraph 36 (Page 17) of NEPRA's CPPA-G CY 2026 Power Purchase Price Forecast Decision dated 07-01-2026 and re-confirmed verbatim in Paragraph 23 (Page 7) of S.R.O. 1501(I)/2026 dated 07-09-2026:

- **National Average Power Purchase Price (NAPPP)**: **Rs 25.32/kWh** (comprising Rs 8.13/kWh Energy Purchase Price + Rs 16.05/kWh Capacity Purchase Price + Rs 1.14/kWh Transmission/Market Operator

charges at the wholesale NTDC level).

- **National Average Energy Purchase Price (NAEPP)**: **Rs 8.13/kWh** (the pure wholesale fuel + variable O&M energy cost at the CPPA-G pool level, which rises to **Rs 9.12/kWh** at the consumer distribution level after adjusting for 11.27% T&D losses, and stood near Rs 10–11/kWh under earlier FY 2024-25 pool estimates cited informally to journalists in February 2026).
- In Paragraph 23 of S.R.O. 1501(I)/2026 dated 07-09-2026, NEPRA left zero ambiguity by ruling explicitly: *"Regarding adjustments of Net Metering / Net Billing units, the Authority had already determined the National Average Power Purchase Price (NAPPP) as Rs.25.32/kWh and National Average Energy Purchase Price (NAEPP) as Rs.8.13/kWh in the PPP decision dated 07.01.2026."* Thus, **Rs 8.13/kWh** is the legally binding NAEPP buyback rate for post-9 February 2026 prosumers in CY 2026.

2. Grandfathering Cutoff Date, Contract Duration, Renewal, and Legal Challenges

When S.R.O. 251(I)/2026 was originally published on 9 February 2026, Regulation 21(2) preserved the old Rs 25.32/kWh rate until contract expiry *only* for quarterly net excess cash-outs while subjecting existing solar users' monthly billing to Regulation 14 (net billing). Following an immediate public outcry, a directive from Prime Minister Shehbaz Sharif on 12 February 2026 to protect existing ~466,000 solar users, and a writ petition before the **Lahore High Court (LHC)** heard by Justice Abid Hussain Chattah on 14 February 2026 (Dawn, 15-02-2026), NEPRA published a draft amendment on 20 February 2026 and notified the final **First Amendment via S.R.O. 547(I)/2026 on 02-04-2026**:

- **Cutoff Date**: Agreements validly executed **prior to 9 February 2026** (< `09-02-2026`) are fully grandfathered because S.R.O. 547(I)/2026 Section 1(2) states: *"They shall come into force at once and shall be deemed to have taken effect on and from the 9th day of February, 2026."*
- **Grandfathered Rights & Contract Duration**: Existing net-metering prosumers retain both **1:1 kWh monthly unit netting** against Off-Peak imports and the **Rs 25.32/kWh (NAPPP)** buyback rate until the expiry of their existing **7-year agreement term**.
- **Loss of Grandfathering via Modification or Renewal**: Under the proviso to amended Regulation 21(2) (S.R.O. 547(I)/2026), grandfathering applies only so long as the agreement is **not modified or altered in any substantive manner that changes the maximum electrical output** of the distributed generation system. Expanding inverter/panel capacity or renewing the contract upon the expiry of the 7-year term (under Regulation 21(3)) automatically transitions the consumer to a **5-year Net-Billing agreement** at NAEPP (**Rs 8.13/kWh**).

Topic E: Meter Types in Lahore Residences (LESCO)

Verification Matrix of LESCO Residential Meter Types and Smart Meter Rollout

LESCO's residential metering infrastructure in Lahore is in the midst of a major transition from legacy unidirectional static and bidirectional "green" meters to Advanced Metering Infrastructure (AMI) smart meters. The table below summarizes the verified meter classifications, "green meter" terminology, rollout statistics, and meter requirements across primary NEPRA/Ministry of Energy records and secondary utility/news sources (LESCO Rebasing Decision 07-01-2026, S.R.O. 251(I)/2026, Business Recorder, 25-08-2026, The Express Tribune, Sep 2025 / Aug 2026, ProPakistani, Sep 2026, and eBillPakistan).

Meter Topic / Type	Verified Status, Technical Role & LESCO Statistics	Effective / Reference Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
1. Single-Phase Static Digital	Standard unidirectional electronic kWh meter installed on residential	Active in CY 2026	LESCO Rebasing	ProPakistani (Sep 2026);	High

Meter Topic / Type	Verified Status, Technical Role & LESCO Statistics	Effective / Reference Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
Meter (230V)	connections **< 5 kW** (Tariff A-1(a) Lifeline, Protected, Unprotected). Read manually via optical/handheld unit and mobile image capture. Being superseded on new connections by single-phase AMI meters from Sep 2026.	(297,000 single-phase AMI procurement initiated Sep 2026)	Decision 07-01-2026, Annex-V	eBillPakistan; BijliBills	
2. Electromechanical Meter (Legacy Ferraris Disk)	Obsolete induction-disk meter. Phased out for all new connections and systematically replaced under LESCO defective/aging meter replacement drives. Exact remaining residual count in older Lahore subdivisions is **UNVERIFIED**.	Legacy / Phased Out	LESCO Rebasing Decision 07-01-2026	eBillPakistan; BijliBills; SolarWorks	High (Residual Share UNVERIFIED)
3. Three-Phase Static ToU Digital Meter (400V)	Multi-register electronic meter recording Peak kWh, Off-Peak kWh, and Maximum Demand Indicator (MDI in kW). Mandatory for all residential connections **≥ 5 kW** (Tariff A-1(b)).	Active in CY 2026	LESCO Rebasing Decision 07-01-2026, Annex-V Part-I Clause 11	eBillPakistan; LESCO Online Bills; PakEra	High
4. Bidirectional / "Green Meter" vs. AMI Smart Meter	**"Green Meter"** is LESCO's colloquial & administrative term for a **3-phase bidirectional digital meter** (often housed in a green/polycarbonate body) used for solar net metering up to Aug 2025. In **Sep 2025**, LESCO halted "green meter" procurement and mandated **3-phase AMI bidirectional smart meters** for all new solar prosumers.	Sep 2025 transition; codified in S.R.O. 251(I)/2026 Reg. 12	S.R.O. 251(I)/2026 Reg. 12	Express Tribune (Sep 2025); ProPakistani; SolarWorks	High
5. AMI / AMR Smart Meter Rollout Status in Lahore	**~1,150,000 (1.15 million) smart meters installed in LESCO** out of >2.1 million nationwide (reported by Ministry of Energy to Senate on 25 Aug 2026). LESCO ordered **297,000 single-phase AMI meters** in Sep 2026 (new 1-phase AMI demand notice: ~Rs 17,700–21,300). Exact % share by meter type across LESCO's ~6.5M total consumers is **UNVERIFIED**.	25 Aug 2026 (Senate report) & Sep 2026 (1-phase AMI rollout)	Ministry of Energy Senate Reply / NEPRA LESCO Capex Filings	Business Recorder (25-08-2026); ProPakistani (Sep 2026); Express Tribune	High (Exact % Share UNVERIFIED)
6. Pre-Paid Residential Meter (Category A-1(c))	Tariff Category A-1(c) is notified at **Rs 42.12/kWh** variable + **Rs 675/kW/month** fixed charge (S.R.O. 279(I)/2026), but actual residential physical prepaid card/token meter deployment in Lahore is negligible / **UNVERIFIED**.	12 Feb 2026 (Tariff rate)	NEPRA Decision 11-02-2026; S.R.O. 1161/2025	Express Tribune; eBillPakistan; BijliBills	High (Deployment Count UNVERIFIED)

What "Green Meter" Actually Means in LESCO Usage and Source Disagreements

In Lahore's residential solar market, the term **"green meter"** has two overlapping meanings that occasionally cause confusion between consumers, solar installers, and utility engineers:

1. **Primary LESCO & Consumer Meaning (Solar Net-Metering Bidirectional Meter)**: Historically, when a Lahore resident applied for solar net metering under the 2015 Regulations, LESCO required replacing the unidirectional three-phase ToU meter with a specialized **three-phase four-quadrant bidirectional static digital meter** capable of recording four distinct energy registers: `Import Peak`, `Import Off-Peak`, `Export Peak`, and `Export Off-Peak`. Because LESCO's approved vendor meters for net metering featured a distinctive green faceplate/enclosure and served renewable ("green") solar energy, both LESCO sub-divisional offices and consumers universally referred to any net-metering bidirectional meter as a **"green meter"** (The Express Tribune, Sep 2025; SolarWorks, 2026).
2. **Why Some Sources Say "Green Meters Are Banned/Discontinued"**: In **September 2025**, LESCO formally halted the procurement and installation of standalone non-communicating bidirectional "green meters" and replaced them with **three-phase AMI (Advanced Metering Infrastructure) smart meters** (priced around Rs 42,000–45,000 on solar demand notices). An AMI smart meter is inherently a **three-phase, Time-of-Use, bidirectional net-metering/net-billing meter** equipped with an embedded cellular (GPRS/4G) or PLC communication module that transmits real-time import/export registers to LESCO's central data center without human meter readers. Thus, while colloquial usage still calls any solar export meter a "green meter," technically LESCO now installs **3-phase AMI bidirectional smart meters** for all new prosumer connections.
3. **Relation to Phase, ToU, and Net Metering**: Because both the 2015 Net Metering Regulations and Regulation 2(1)(v) of the NEPRA (Prosumer) Regulations, 2026 (S.R.O. 251(I)/2026) restrict prosumer eligibility to **three-phase (400V or 11kV)** connections, every residential net-metering/net-billing meter in LESCO is simultaneously:
    - **Three-Phase (400V)**,
    - **Time-of-Use (ToU)** (recording Peak and Off-Peak windows), and
    - **Bidirectional** (recording imported and exported kWh separately).

Conversely, a consumer who requires **only ToU billing** (≥5 kW load without solar) needs a **three-phase ToU digital or AMI meter** (which does not have export billing enabled unless a Prosumer Agreement is executed), whereas a consumer who requires **net metering / net billing** must have a **three-phase connection with a LESCO-approved bidirectional AMI smart meter** (or legacy bidirectional "green meter" if commissioned before September 2025, or two separate unidirectional meters under Regulation 12(1) of S.R.O. 251(I)/2026).

Topic F: Change and Risk Check (Next 12 Months: November 2026 – October 2027)

Verification Matrix of Upcoming Tariff Changes, Policy Reforms, and Litigation

For product configuration and financial modeling of residential solar-plus-storage systems over the next 12 months (November 2026 through October 2027), six material regulatory, tariff, and judicial developments have been formally initiated or filed. The table below documents each risk item across primary NEPRA admission notices/orders and independent secondary reporting (LESCO CY 2027 Tariff Petition Admission Notice, Sep 2026, NEPRA September 2026 FCA Notice, Dawn, 15-02-2026, Dawn, 23-09-2026, Business Recorder, Oct 2026, and The Express Tribune).

Change / Risk Item	Verified Scope, Financial Magnitude & Timeline	Effective / Target Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
1. CY 2027 Annual Tariff Rebasing (LESCO & CPPA-G Petitions)	All 10 XWDISCOs filed CY 2027 petitions in Sep 2026. **LESCO requests Rs 79,274M net distribution margin + Rs 32,782M Prior Year Adjustment (PYA)** = **Rs 31.71/kWh** average cost-of-service rate (11.27% T&D loss target on 30,275 GWh sales). Nationally, DISCOs seek **Rs 268 billion in PYA**, while CPPA-G projects CY 2027 PPP at **Rs 27.309/kWh** (slight drop of Rs 0.65–1.37/kWh).	**01 Jan 2027** (statutory rebasing window: 1–15 Jan 2027)	NEPRA LESCO CY 2027 Petition Admission Notice (Sep 2026)	Business Recorder; Dawn; Express Tribune	High
2. November 2026 FCA Spike (+Rs 2.5624/kWh Requested)	CPPA-G filed a petition seeking **+Rs 2.5624/kWh** positive FCA for **September 2026** consumption, with public hearing scheduled on **29 October 2026**, to be recovered in **November 2026** bills.	**Nov 2026 Bills** (Hearing: 29 Oct 2026)	NEPRA FCA Hearing Notices (Oct 2026)	Business Recorder; Dawn; Qaaf	High
3. IMF-Backed BISP Direct Cash Subsidy Replacement for ≤200-Unit Consumers	Under IMF EFF structural benchmarks and National Electricity Plan Strategic Directive 067, the Federal Govt plans to replace blanket tariff subsidies for Protected (≤200-unit) consumers with **direct BISP-linked targeted cash transfers / CNIC verification** starting **January 2027**.	**Jan 2027** (Target Launch)	Power Division / NEPRA Tariff Rationalization Record	Dawn; Business Recorder; Express Tribune	High
4. Power Division Proposal to Relax 6-Month Protected Penalty	In Sep 2026, Power Division proposed amending the 6-month consecutive ≤200-unit rule so that a Protected consumer exceeding 200 units in a single summer month pays the Unprotected rate **only for that specific month** rather than being penalized for 6 subsequent months. Currently at **proposal/review stage** (not yet notified).	Pending / Under Review (Sep–Dec 2026)	Power Division Policy Briefings (Sep 2026)	Dawn (23-09-2026); BijliBills; eBillPakistan	High (Enactment Uncertain)
5. Pending NEPRA Review Motion on Per-kW Residential Fixed Charges	Formal review motion filed in March 2026 (led by former Interior Minister Lt Gen (retd) Moin ud Din Haider and consumer interveners) challenging the **11 Feb 2026 / S.R.O. 279(I)/2026** imposition of **Rs 200–675/kW/month** fixed charges on Protected and Unprotected residential consumers.	Pending NEPRA Review Decision	NEPRA Tariff Rationalization Proceedings	Dawn; Business Recorder; Express Tribune	High
6. Pending High Court Petitions (LHC & IHC) Against Prosumer Regulations 2026	Constitutional writ petitions pending in **Lahore High Court (LHC)** (filed Feb 2026; Justice Abid Hussain Chattah issued notices to GoP, NEPRA, and LESCO without granting immediate stay) and **Islamabad High Court (IHC)** (filed Apr 2026 by RIAA Barker	Ongoing Judicial Proceedings (2026–2027)	S.R.O. 251(I)/2026 & S.R.O. 547(I)/2026 Record	Dawn (15-02-2026); RIAA Barker Gillette (Apr 2026); Express Tribune	High

Change / Risk Item	Verified Scope, Financial Magnitude & Timeline	Effective / Target Date	Primary Source URL	Secondary Sources (≥3 Independent)	Confidence
	Gillette) challenging Net Billing and the Rs 8.13/kWh buyback rate for new prosumers.				

Conclusions and Solar/Battery Energy-Management System (EMS) Configuration Synthesis

To configure a residential solar and battery energy-management product in Lahore accurately, the software engine must decouple consumer profiles into three distinct optimization regimes:

1. **Post-9 February 2026 Three-Phase Prosumers (Net-Billing Regime under S.R.O. 251(I)/2026)**:

   • **Self-Consumption Priority over Grid Export**: Because daytime grid exports earn only **Rs 8.13/kWh** (NAEPP) with zero tax or surcharge offset while grid imports cost **Rs 45.77/kWh** all-in during Off-Peak hours (**Rs [34.53 + 3.23 + 0.5194 + 1.1086] × 1.18** before ED) and **Rs 60.31/kWh** all-in during Peak hours (**Rs [46.85 + 3.23 + 0.5194 + 1.1086] × 1.18** before ED), every kilowatt-hour of solar generation stored in a home battery and discharged during the evening Peak window saves **Rs 52.18/kWh net** compared to exporting that unit at noon.

   • **Automated Seasonal Peak-Window Switching**: The EMS firmware must automatically shift its 4-hour evening battery discharge schedule across the four statutory LESCO seasons: **17:00–21:00** (Dec–Feb), **18:00–22:00** (Mar–May), **19:00–23:00** (Jun–Aug), and **18:00–22:00** (Sep–Nov).

2. **Pre-9 February 2026 Grandfathered Three-Phase Prosumers (Legacy Net-Metering Regime under S.R.O. 547(I)/2026)**:

   • Until the expiry of their 7-year contract term, grandfathered users retain **1:1 kWh monthly netting** between daytime solar exports and Off-Peak grid imports, plus **Rs 25.32/kWh (NAPPP)** on quarterly net surplus units. For these users, the battery need only cover the **4-hour Peak window** (since Peak imports cannot be offset 1:1 by daytime Off-Peak exports) and grid outages, while daytime excess solar can continue to bank 1:1 against nighttime Off-Peak air-conditioning loads.

3. **Single-Phase Non-ToU Slab Consumers (< 5 kW, Zero-Export / Hybrid Self-Consumption)**:

   • Because single-phase connections (< 5 kW) are ineligible for a prosumer net-billing agreement without upgrading to a 3-phase meter, single-phase hybrid solar-plus-battery systems must operate with **hard zero-export limiting** (to prevent unidirectional digital or AMI meters from billing reverse current as forward consumption).

   • For single-phase consumers near the **200 kWh/month Protected threshold**, the EMS should implement a **Cumulative Monthly Grid-Import Guardrail**: keeping monthly grid import at or below **195 kWh** preserves the Protected tariff (**Rs 10.54–13.01/kWh** + Rs 0.43/kWh FC surcharge) and avoids triggering a six-month penalty into the Unprotected tariff (**Rs 28.91–33.10/kWh** + Rs 3.23/kWh FC surcharge). Similarly, for Unprotected consumers, shaving monthly grid imports just below whole-bill slab boundaries (100, 200, 300, 400, 500, 600, and 700 kWh) captures massive step-function bill reductions across the entire month's consumption.

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